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REGULAR PAPER

A novel sentiment correlation-based method with dual transformer model for stock price prediction

Qizhao Chen[1] · **Hiroaki Kawashima**[1]

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Abstract
In real-world financial markets, a company's stock performance is often influenced by its competitors, partners, and other connected firms. However, traditional sentiment-based approaches typically focus only on the target company's sentiment, overlooking the broader network of influence. To address this limitation, this paper proposes a Dual Transformer architecture that integrates sentiment signals from connected companies to enhance prediction accuracy. This approach aims to bridge a critical gap in financial sentiment analysis by capturing inter-company relationships and their impact on stock price movements. The Dual Transformer comprises two transformer structures, the Enhancement Transformer and the Forecast Transformer. The Enhancement Transformer is used to enhance the correlation strengths between related companies, while the Forecast Transformer is used for stock price forecasting. Eight companies from various industries and financial markets are selected for analysis. The model utilizes the polarity scores of these related companies, combined with historical closing prices from 2015 to 2024, to forecast the next day's closing price. Prediction performance is evaluated using the mean squared error (MSE). Experimental results demonstrate that incorporating the news sentiment of related companies improves the prediction of the target company's stock price when using the proposed Dual Transformer model. In addition, the Dual Transformer model can outperform existing models such as Temporal Fusion Transformer, N-Beats, Informer, and LSTM by achieving consistently lower MSE values across all eight companies.

Keywords Time series forecasting · Sentiment analysis · Sentiment correlation · Transformer · Named entity recognition

1 Introduction

Stock price prediction has long been a topic of great interest for financial analysts, investors, and researchers. Conventionally, these forecasts are based on historical stock performance, financial statements, and macroeconomic indicators [1]. However, with the increasing availability of textual data, particularly news articles, sentiment analysis has become a valuable method for stock price prediction. The sentiments embedded in news articles often provide important information on market perceptions and investor behavior, which can affect stock movement [2].

Although recent studies have focused on investigating the sentiment in the news directly related to a target company [3–8], some research suggests that strong media sentiment toward a company, often caused by a significant real-world event, could have a spillover effect on related companies [9]. Therefore, the stock price of a company can also be affected by news related to its connected companies, such as competitors in the same industry or sector. In reality, no company can operate in isolation, and it forms a broader network of suppliers, customers, competitors, and business partners. Events that influence one company can also have an impact on others within the network, which can be used to predict the stock price. However, traditional sentiment-based approaches typically consider only the sentiment of the target company [10]. Furthermore, effective capture of inter-company sentiment relationships remains a critical challenge in the field.

Furthermore, incorporating sentiment analysis into stock price prediction models can significantly improve the precision of prediction [11–13]. However, effectively leveraging news sentiment poses challenges, particularly when there is

✉ Qizhao Chen
af24o008@guh.u-hyogo.ac.jp

Hiroaki Kawashima
kawashima@gsis.u-hyogo.ac.jp

[1] Graduate School of Information Science, University of Hyogo, Kobe 651-2197, Japan

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minimal news coverage about target companies, as is often the case with stocks that have low trading volumes. In such situations, it becomes essential to consider alternative data sources, such as news about companies that are related to or have significant business ties to the target firms. Thus, exploring these indirect sentiment sources and developing methods to integrate them into predictive models is crucial to improving the reliability of forecasts.

Moreover, research on sentiment-driven stock price prediction and sentiment linkages between different companies has significant real-world implications. For example, in portfolio management, identifying sentiment linkages between companies can improve risk assessment and inform diversification strategies, reducing overexposure to highly correlated assets. In risk management, tracking sentiment among related companies can act as an early warning system for sector-wide or market-wide risks, helping mitigate potential losses.

When it comes to prediction models in stock price forecasting, traditionally, some frequently used models include recurrent neural network (RNN) (e.g., long short-term memory (LSTM) and bidirectional LSTM (Bi-LSTM)) [14–16], convolutional neural network (CNN) [17, 18], support vector machine (SVM) [19–22] and ensemble methods such as Random Forest [23]. The limitations of some classical machine learning models, such as SVM, are evident. These models fail to capture the complicated temporal and dynamic relationships within the stock price data. Although RNNs excel at capturing temporal dependencies, they also have limitations in stock price prediction. For example, RNNs struggle with highly volatile or noisy data, which is common in stock prices, potentially leading to overfitting or reduced accuracy.

Currently, due to its outstanding performance in sequential tasks, Transformer becomes a popular candidate in stock price prediction [24–26]. The Transformer model, proposed by [27], is a deep learning architecture widely used in natural language processing (NLP) and other sequential tasks, including time series forecasting. For example, some Transformer-based models have been shown to be able to outperform some classical models such as LSTM in stock price prediction [28–30].

However, most existing Transformer-based models in finance focus solely on the target company's own historical data or sentiment [31], overlooking the influence of other related companies such as competitors, suppliers, or strategic partners. Although Transformers excel at capturing temporal dependencies within a single input sequence, they are not explicitly designed to model inter-company sentiment relationships. This limitation motivates our proposed Dual Transformer architecture, which introduces a two-stage framework: the first Transformer encodes sentiment signals from multiple related companies, and the second integrates this encoded sentiment with historical price data to forecast the target company's stock price.

By decoupling sentiment modeling and price forecasting, the Dual Transformer enables more effective integration of external sentiment influences. This structure differs from existing approaches by explicitly modeling cross-company sentiment interactions, a dimension often ignored in previous Transformer-based financial models. Therefore, our work not only applies Transformer architecture to stock prediction but also extends its capability to capture multi-entity relationships, offering both methodological and empirical advancements.

This paper aims to explore whether the news sentiment of related companies can improve the prediction of the stock price of a target company with the proposed Dual Transformer model. The main contributions of this research are:

1. We propose a framework by incorporating news sentiment of related companies in stock price prediction and demonstrate that news sentiment of related companies offer predictive value for the target company's stock price movement when using the proposed model.
2. We propose a novel Dual Transformer model to increase the correlation strengths between related companies and enhance the performance of stock price prediction. We show that the Dual Transformer model can increase the correlation strength between companies and outperform models such as Informer [32], Temporal Fusion Transformer [33], N-Beats [34], and LSTM.
3. We introduce an innovative use of Named Entity Recognition (NER) to identify related companies, complemented by methods such as SHAP (Shapley Additive Explanations) to refine the related companies further.

The remainder of this paper is structured as follows. Firstly, the related work is listed (Sect. 2). The methodology is then described (Sect. 3). In Sect. 4, the experimental results are presented. Finally, we conclude with Sect. 5.

2 Related work

2.1 Sentiment analysis in stock price prediction

The use of sentiment analysis in stock price prediction has attracted growing attention in recent years, driven by the increasing availability of textual data and the rapid development of NLP. This subsection summarizes key contributions in these areas, focusing on news impact on stock price and sentiment analysis on stock price prediction.

There are almost no studies related to sentiment correlation of related companies, which may be due to the fact that the companies chosen in most research are well-known companies locally or globally, such as Google and Microsoft, and there are tons of news or comments about these companies

online. Therefore, alternative data are not needed in these cases. Currently, most of the research still focuses on the application of the sentiments of target companies to their stock price forecasting.

Research on the application of sentiment analysis in stock price forecasting has explored various methods and data sources. Several studies focus on using social media sentiment for prediction. For example, Upadhyay et al. [35] introduced a novel machine learning-based model called TextBlob Hybrid Arima-Garch (TBHAG) for stock prediction. Twitter data is used for sentiment analysis. Das et al. [36] proposed an EEMD–ensemble CNN model that combines decomposed stock price signals, technical indicators, and Twitter (X) sentiment scores to improve stock price prediction accuracy. By integrating public sentiment through a dedicated CNN and leveraging multiple IMF-based subnetworks, the model outperforms baseline methods and highlights the value of social media data in financial forecasting. Liu [37] proposed a multimodal factor mining approach that combines market data, financial texts, and social media sentiment to enhance stock market forecasting accuracy and interpretability. Batra, Rakhi, and Daudpota [21] applied SVM to predict the sentiment of tweets about Apple from StockTwits, demonstrating a positive relationship between sentiments and stock price. Similarly, Nguyen, Thien, and Shirai [20] proposed Topic Sentiment Latent Dirichlet Allocation (TSLDA) to analyze social media sentiments, finding that incorporating sentiment data improves stock price predictions. Pagolu et al. [23] also analyzed tweets, combining human-annotated and machine-classified sentiments, and observed a strong correlation between Twitter sentiment and stock price movements.

Other research has explored the sentiment of financial news and its predictive power. Khedr, Ayman, and Yaseen [38] used Naive Bayes to extract sentiment polarity from financial news and combined it with historical stock prices, achieving an accuracy of up to 89.80%. Mehta, Pandya, and Kotecha [19] examined how public opinions in financial news influence stock prices. They used machine learning models, including Naive Bayes, Maximum Entropy (ME), SVM, and MLP, and found that LSTM was particularly effective for stock price prediction. Lee et al. [39] introduced a deep learning approach that integrates ESG sentiment indices from news with technical indicators to predict the S&P 500 index. Liu et al. [40] proposed SA-TrellisNet, a news-driven stock index prediction model that combines a CNN-LSTM-based sentiment analysis module with a sentiment attention mechanism and TrellisNet for accurate forecasting. Kirtac and Germano [41] evaluated the effectiveness of LLMs such as OPT, BERT, and FinBERT in analyzing financial news sentiment, finding that the GPT-3-based OPT model achieves 74.4% accuracy in predicting stock returns. A trading strategy using OPT yields a Sharpe ratio of 3.05 and a 355% return over two years, highlighting LLMs' potential in market forecasting and portfolio management. Mo et al. [42] applied sentiment analysis to financial news headlines using fine-tuned large language models, finding that the gemma-7b model achieves the best performance in classifying investor sentiment.

A comparative study by Li et al. [22] evaluated the performance of sentiment analysis against other approaches such as bag-of-words and sentiment polarity. They found that sentiment analysis models outperformed bag-of-words approaches at stock, sector, and index levels, while sentiment polarity models offered limited predictive value.

In terms of hybrid approaches, Srijiranon et al. [43] proposed a PCA-EMD-LSTM model that incorporates news sentiment and technical indicators to predict the next day's stock price. Their findings revealed that integrating news sentiment analysis significantly enhances the performance of LSTM-based models.

2.2 Transformer-based models in time series forecasting

Recent advances in stock price prediction have leveraged Transformer-based architectures and specialized time series models to improve forecasting performance. Informer [32] introduces a sparse self-attention mechanism and a generative decoder, enabling efficient long-sequence forecasting. Autoformer [44] further improves long-term prediction by incorporating a series decomposition block and an autocorrelation mechanism to capture seasonal and trend components more effectively. Temporal Fusion Transformer (TFT) [33] integrates attention with static covariates and gating mechanisms for interpretability and multi-horizon forecasting, while N-BEATS [34] uses backward and forward residual links in a deep MLP architecture and performs strongly on benchmark datasets. Although these models achieve competitive performance in structured time series prediction, they are not designed to integrate unstructured financial news. Some recent efforts [45] attempt to combine text and price signals but do not explicitly capture inter-company sentiment spillovers. Our proposed Dual Transformer model addresses this gap through a two-stage architecture: the first Transformer encodes sentiment relationships across firms, and the second integrates this representation with historical price data for forecasting. This design enables richer modeling of multi-firm sentiment dynamics alongside temporal dependencies.

3 Methodology

Figure 1 provides an overview of the research setup presented in this paper.

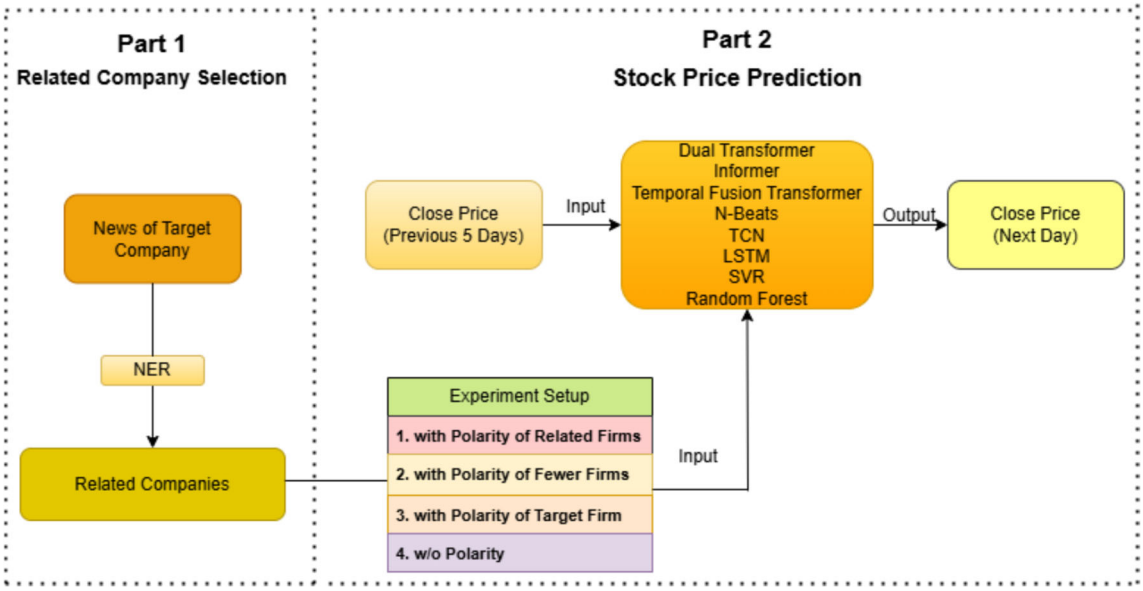


Fig. 1 Full Picture of Research Setup

There are basically two parts to perform the analysis. Part 1 is the related company selection. The definition of a related company is broad. However, in this research paper, we define the related companies as the companies which frequently co-occur in news articles concerning the target company because if a company frequently appears in news articles about the target company, it is likely that the two companies are connected. By applying Named Entity Recognition (NER), we can extract the companies from the news articles of the target company. The proposed methods for identifying related companies are more user-friendly for individuals with limited knowledge of finance. Unlike traditional fundamental analysis, such as reviewing financial statements, which can be time-consuming, our method offers a more efficient alternative. Furthermore, if users are unfamiliar with certain companies, especially those from other countries, traditional analysis may require significant time and effort, making our approach even more advantageous.

Part 2 is the stock price prediction. We mainly use two types of feature to forecast the stock price. The first type is historical close price, and the other type is polarity score of companies. We propose a Dual Transformer model to capture the interaction between companies and to forecast the stock price. To evaluate the effectiveness of this Dual Transformer model, we compare its performance against several deep learning models, such as Informer, and traditional machine learning models, including Random Forest. In addition, to investigate whether the number of companies used in the prediction affects the results, we design an experiment featuring various scenarios. The details of this experiment will be discussed in Sect. 3.6.

3.1 Data

There are two types of data used in this research, news data and stock price data. The financial news data of companies were downloaded using the News Feed and News Sentiment data API [1] from websites such as Yahoo News, InvestorPlace and GlobeNewswire. The collected data table mainly include the columns "date", "title", "content", "link", "symbols", and "sentiment". The "sentiment" column contains the polarity score of the corresponding news, which is calculated based on negative and positive mentions in texts normalized with possible values between −1 and 1. In sentiment analysis, the polarity score quantifies the sentiment expressed in a text and ranges from −1 to 1, where negative values represent negative sentiment, positive values represent positive sentiment and values close to 0 suggest neutral sentiment. Each selected company has about 4000 news articles. Since multiple pieces of news are released on some days, we use the average polarity score for each day to represent the overall sentiment of that day. These polarity scores will serve as input for predicting stock prices in this research paper.

For stock price data, the Python library yfinance helps import the stock price of companies from Yahoo Finance. The imported data mainly contain five features, "Open", "Close", "High", "Low", and "Volume". Only the close price ("Close"), the stock price at the end of each trading day, is used for prediction because it is often considered to be the most important price of the day and it is the final consensus of the stock value among traders and investors. Based on related research in stock price prediction [6, 7, 11, 12, 46], the

[1] https://eodhd.com/financial-apis/stock-market-financial-news-api

stock prices of four to ten companies are typically included in the analysis. In this paper, eight companies' stock price from 01/05/2015 to 08/05/2024 is used, which include Apple (AAPL), Airbus (EADSY), HSBC (0005.HK), InfuSystem (INFU), Toyota (7203.T), Tencent (0700.HK), Pepsi (PEP), and Petrobras (PBR).

The selected companies represent diverse financial markets and sectors. Apple is an American company renowned for its innovative software and electronic products. Airbus is a leading European aerospace company that designs, manufactures, and delivers commercial aircraft, helicopters, and defense systems. HSBC is a British banking and financial services company headquartered in London. InfuSystem Holdings is an American healthcare company specializing in integrated therapy services and medical equipment management. Toyota, based in Japan, is one of the largest automotive manufacturers in the world. Tencent, a technology company headquartered in Shenzhen, China, owns business in social networks, gaming, and entertainment. PepsiCo is an American company with a strong presence in the food and beverage industry. Petrobras is a Brazilian state-controlled energy company primarily engaged in oil and gas exploration, production, refining, and distribution.

Historical close prices and polarity scores, using a sliding window of 5 days, will be utilized to predict the close price for the following day. 70% of the data (2015–2021) will be used for training and 30% of the data (2021–2024) will be used for testing. The MinMax Scaler will be applied to scale the feature values between 0 and 1 to avoid features with larger values dominating the prediction.

To address potential information leakage, we ensure that only news published on the previous day or earlier is used to predict the next day's stock price, thus preventing any look-ahead bias.

To ensure consistency across all firms, all news articles are processed in Coordinated Universal Time (UTC), providing a uniform time reference independent of local time zones. This ensures that news published at the same UTC date is consistently assigned to that day, regardless of the company's geographic location. By using UTC, we avoid issues caused by time zone differences and daylight saving time changes, maintaining a globally consistent calendar day for sentiment aggregation. This approach enables fair and accurate comparison of news sentiment across companies traded in diverse markets worldwide.

3.2 Methods to identify related companies

In NLP, NER involves identifying and classifying entities such as organizations, people, dates, and other specific terms within a text. Through entity tagging, NER helps structure unstructured data for better data analysis. In Python, the NLP library spaCy makes it easy to perform NER tasks and the inputs are news articles of companies. In this paper, as shown in Fig. 2, the entity names are extracted from news articles. The companies are then selected from the list of entity names based on their frequency of appearance in news articles, ranked from highest to lowest.

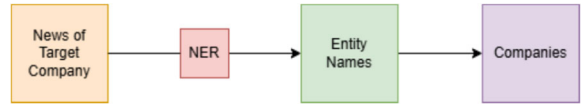


Fig. 2 NER Procedure

Furthermore, in one of our experimental settings in the prediction of stock prices, we need to reduce the number of related companies. Here, we will use the Shapley value to measure the feature importance of the polarity scores from different companies and choose only the top five with the highest scores.

The Shapley value (Equation 1), which originates from cooperative game theory, is used to measure the contribution of players in a coalition. Shapley value of a feature is its contribution to the payout. For example, in machine learning, the "players" are the features and the "payout" is the model's prediction [47]. SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee in 2016 [48], is a framework and methodology to quantify the contributions of the features of machine learning models based on the idea of Shapley value. The Python implementation of SHAP is available. In this research, the SHAP values are generated using the Random Forest with SHAP's TreeExplainer in Python. This choice is based on experimental results that show minimal differences in feature rankings derived from SHAP values between different models.

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (1)$$

where ϕ_i is the Shapley value for the feature i, S is a subset of features that do not include i, N is the set of all features, $|S|$ is the size of the subset S, $f(S)$ is the prediction of the model using only the features in subset S, and $f(S \cup \{i\})$ is the prediction of the model using the features in subset S plus the feature i.

3.3 Prediction model

Inspired by [49], where a Dual-path Actor Interaction (DualAI) framework is proposed to model group activities in sports, in this paper, we propose a Dual Transformer model to model interaction between companies and predict stock prices. As the name suggests, this model connects two transformers (Fig. 3), the Enhancement Transformer (**ET**) and the Forecast Transformer (**FT**). The reason for using this

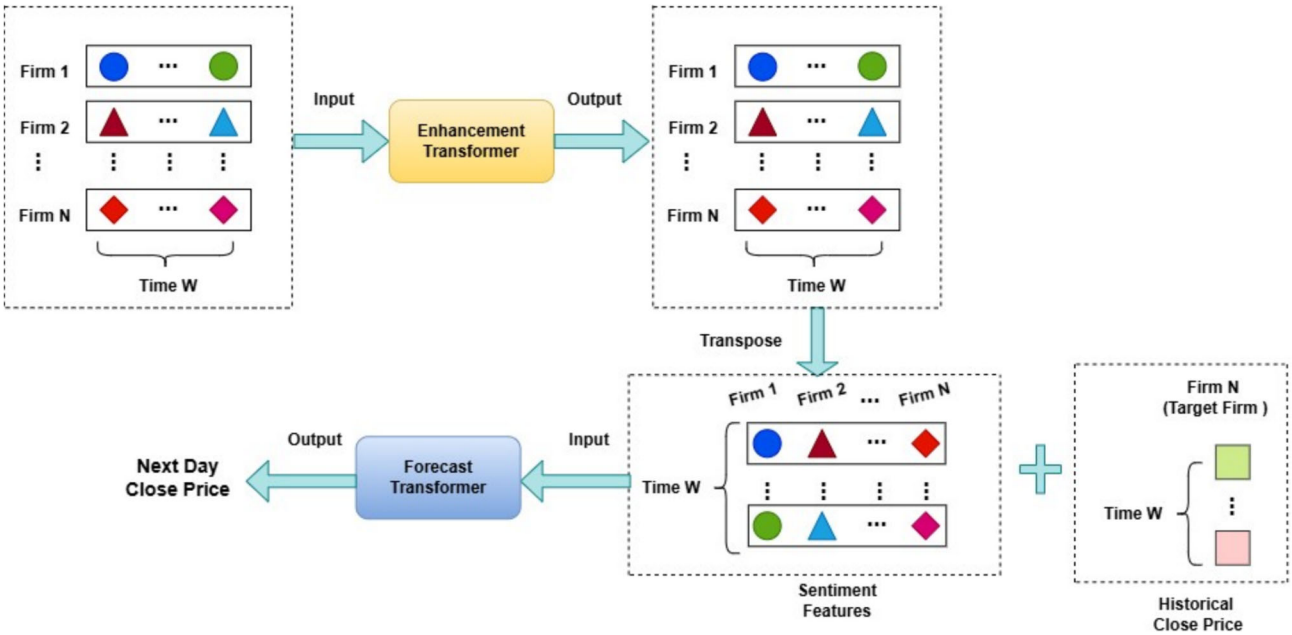


Fig. 3 Dual Transformer

model structure is to effectively process both feature extraction and prediction tasks. The ET focuses on capturing key relationships and patterns within the news sentiment. The FT focuses on mapping the refined features of the ET to accurate stock price predictions. Generally speaking, this model structure separates the responsibilities of feature extraction and prediction, allowing each transformer to specialize in its respective task.

As shown in Fig. 4, the ET consists only of an encoder (without positional encoding), while FT (Fig. 5) includes both an encoder and a decoder, similar to the vanilla Transformer model [27].

3.3.1 Enhancement transformer

The ET takes in the news sentiment of companies as input, and output sentiment features with the same dimensions as the input data. The ET is trained as an autoencoder, as explained later in Sect. 3.4.

To find an optimized way to feed the data to the ET, we investigated how the input configurations of the Enhancement Transformer impact its performance in extracting meaningful features from company polarity scores. Specifically, letting W be the sliding window size, N be the number of companies, and a tuple (d, n) denote the input shape to the model, where d is the dimension of an input vector and n is the number of input vectors to the Transformer, we tested two different input configurations: $(d, n) = (W, N)$ and $(d, n) = (N, W)$.

In the first configuration $(d, n) = (W, N)$, as illustrated in Fig. 4, each input vector consists of the polarity scores of one company over a given time period W. Before passing this into the attention layers, we first project the sliding window size W into a hidden representation of size $d_{model} = 32$, resulting in an input shape of (d_{model}, N).

This approach enables the model to emphasize inter-company dependencies, capturing how sentiment trends across different companies interact and influence one another. This is particularly relevant for industries where companies exhibit strong correlations due to market conditions, sector-wide trends, or shared economic factors.

For the second configuration $(d, n) = (N, W)$, the input is structured such that the polarity scores of different N companies are combined as an input vector and sequential W input vectors are prepared across the time window W. In this case, we project N dimensions into hidden size d_{model}, transforming the input into (d_{model}, W) before feeding it into the attention layers.

This transformation allows the model to focus on capturing temporal dependencies within each company's sentiment trends while preserving company-wise feature relationships. The Transformer then processes this representation to extract meaningful patterns in sentiment evolution and its potential impact on stock price movements.

In the first configuration, the number of companies N does not need to be fixed in advance when training the model, allowing for greater flexibility. This is because the Transformer operates along the temporal dimension (window size W), treating each company as an independent sequence input.

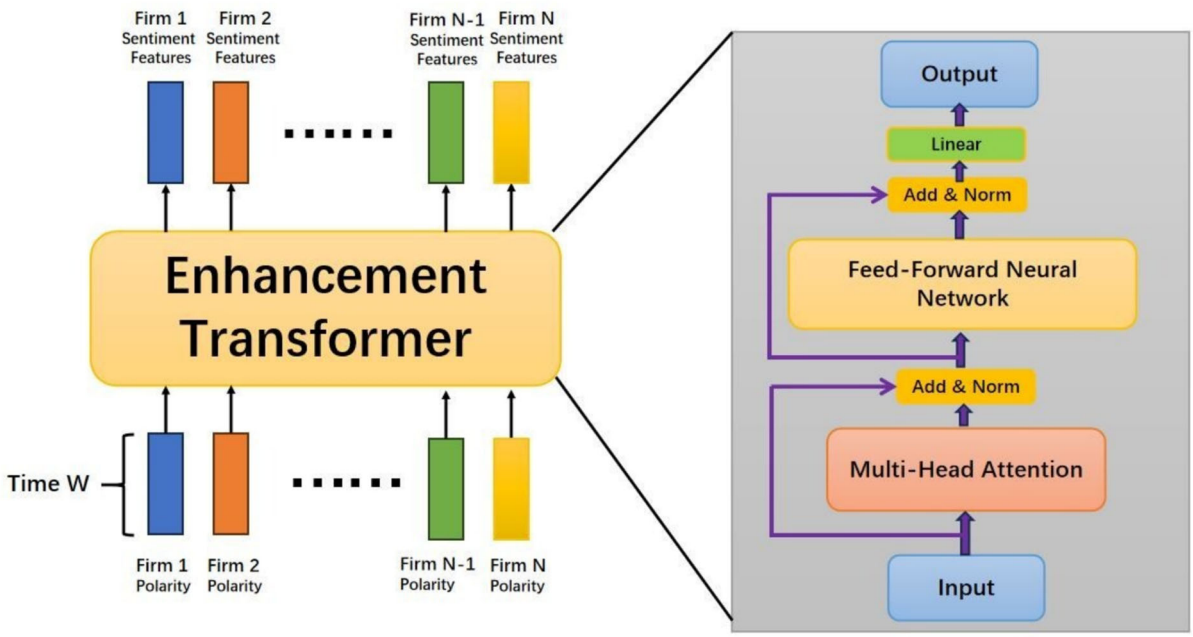


Fig. 4 Enhancement Transformer Structure

Since the model learns to process sequences of a fixed length W, and not across companies, it can flexibly handle varying numbers of companies at inference time. This flexibility is important, as the number of related companies can vary depending on the target company. Moreover, in one of our experimental setups, we intentionally reduce the number of companies to investigate how this affects the models performance.

Additionally, this setup enhances the explainability of the model, as the goal of the Enhancement Transformer is to capture the inter-company relationships and interactions more effectively. By structuring the input such that each vector represents the polarity scores of a single company across the sliding window, this configuration naturally focuses on learning how different companies influence each other, achieving exactly what the Enhancement Transformer is designed to do.

Based on the reasons mentioned above, we choose to continue with the first configuration. Specifically, we denote $X \in \mathbb{R}^{W \times N}$ as the vectors of N companies with W time steps. ET begins by passing the input X into a multi-head attention mechanism (MultiHeadAttention), which captures the dependencies between different companies H_1 (Equation 2). Since companies have no ordering relation, there is no positional encoding. A residual connection is applied by adding H_1 and X, followed by layer normalization (LayerNorm) to stabilize training and output H_2 (Equation 3). Next, H_2 is processed by a feed-forward neural network (FFNN) to further learn nonlinear representations, resulting in H_3 (Equation 4). Another residual connection is applied by adding H_2 and H_3, followed by layer normalization, producing H_4 (Equation 5). Finally, a linear feed-forward network (Linear) is used to further improve the learning capacity of the companies and output $Y \in \mathbb{R}^{W \times N}$ (Equation 6).

$$H_1 = \text{MultiHeadAttention}(X) \tag{2}$$

$$H_2 = \text{LayerNorm}(H_1 + X) \tag{3}$$

$$H_3 = \text{FFNN}(H_2) \tag{4}$$

$$H_4 = \text{LayerNorm}(H_2 + H_3) \tag{5}$$

$$Y = \text{Linear}(H_4) \tag{6}$$

Figure 7 shows the feature correlation before and after ET processing. Figure 7a is the correlation coefficient between polarity scores of companies before ET processing, and Fig. 7b is the correlation coefficient of features output by ET. These two figures were generated using all the training data. Based on the experimental results, after ET processing, the correlation between the sentiment features of different companies is generally stronger, with more cells showing yellow or green shades. This improvement is likely due to ET's ability to capture inter-company interactions, which helps to infer and fill in missing sentiment data for certain companies. This is one of the key motivations behind its design. As the missing data is filled, additional information is introduced, leading to higher correlation coefficients between company sentiment features. The second heat map also shows clearer patterns or clusters of correlation, indicating that the Transformer model may have grouped certain companies based on sentiment similarity.

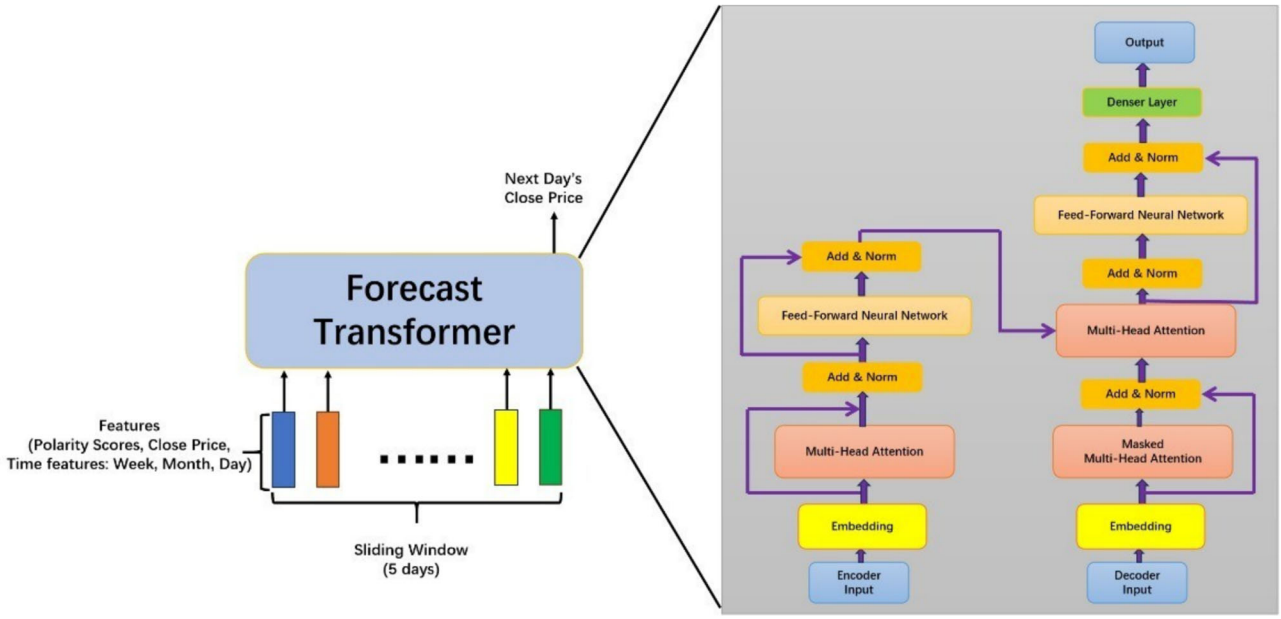


Fig. 5 Forecast Transformer

3.3.2 Forecast transformer

The Forecast Transformer (FT) (Fig. 5) uses an encoder–decoder architecture for stock price prediction and the detailed illustration of the input dimension is shown in Fig. 6. The encoder processes a 5-day sliding window of historical market data that contains three key elements: (1) daily close prices, (2) sentiment features, and (3) time features (day of week, month, and day of month). These inputs are combined through a special embedding layer, where each component is processed separately [32]. The numerical features (prices and polarity scores) pass through a 1D convolutional embedding layer, while the time features are transformed via temporal embedding. All features are then projected into a shared 512-dimensional space and combined with positional information using element-wise summation before being processed by the encoder's attention layers. Experimental results demonstrate that this combined input of price data, sentiment scores, and temporal context yields better predictions than using sentiment scores alone, as the model can learn richer patterns from these complementary information sources.

Note that we use the second input configuration discussed in Sect. 3.3.1 for the FT as opposed to the ET to focus on capturing temporal dependencies. This is because FT is designed to predict the stock price. Therefore, we want the model to capture the evolution of sentiments at different times. Specifically, the input dimension of the encoder is (d_{model}, W), where $W = 5$ represents the size of the sliding window of 5 days.

The decoder in the Forecast Transformer receives two key inputs to predict future stock prices. First, it takes the most recent five-day close prices to create an input sequence like [Day 1 Close, Day 2 Close, Day 3 Close, Day 4 Close, Day 5 Close]. This approach uses the last five known prices as its starting reference points. The number of known prices used for reference is a hyperparameter, defined as pre_len. In the case mentioned above, pre_len is equal to five.

Second, it processes time features (week, month, and day) for all positions in the sliding window sequence. These inputs are transformed through embedding layers: The close price passes through a 1D convolutional layer applied along the time dimension, while the time features are processed by temporal embedding. Both are projected to the same 512-dimensional space and combined with positional encoding before being fed to the decoder layers. This design allows the model to maintain temporal context while focusing on predicting future values, using the last known prices as its starting reference points.

Furthermore, the Forecast Transformer uses a hidden size of $d_{model} = 512$, which has been found to be an effective choice for this task. Reducing d_{model} leads to a degradation in prediction performance, emphasizing the importance of selecting an appropriate hidden size to balance model capacity and computational efficiency.

3.4 Model training

We trained the Enhancement Transformer as an autoencoder using sentiment scores of different companies as inputs. An autoencoder is a type of neural network that learns to copy its input to its output. By training to recreate the input as accurately as possible, the model learns important patterns and

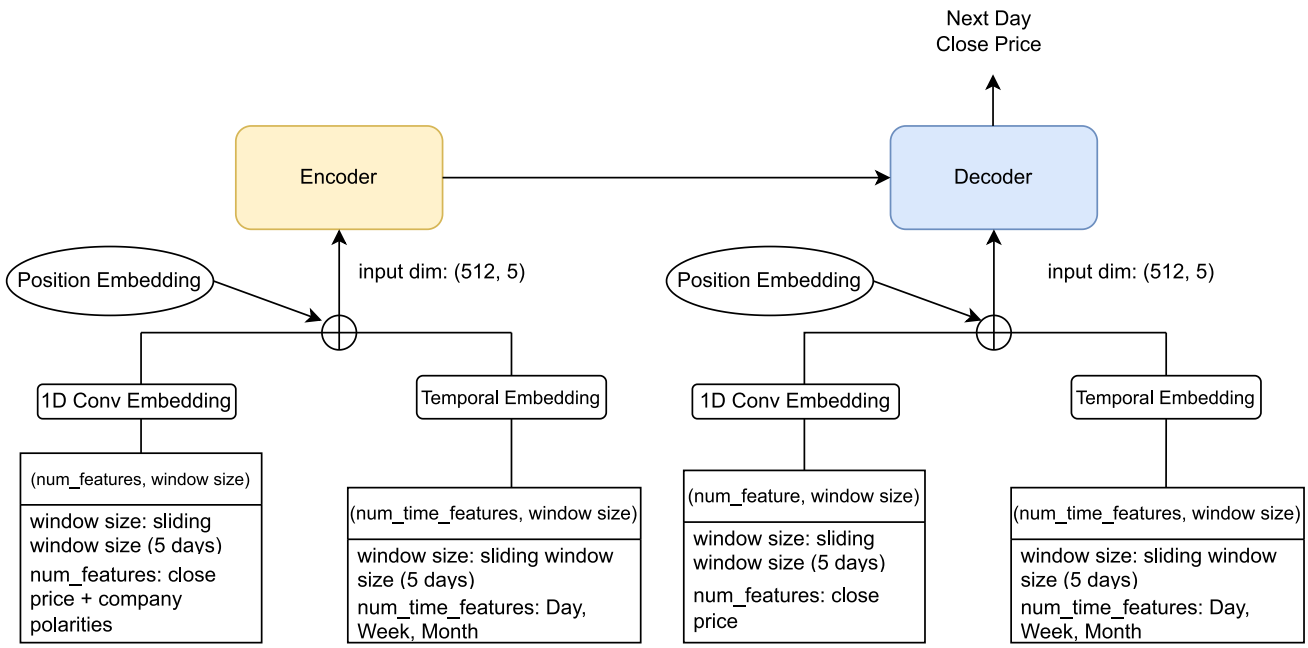


Fig. 6 Forecast Transformer: Detailed Illustration

structures in the data. Autoencoders are often used for tasks such as noise reduction, filling in missing data, or learning compressed features [50].

At each step, the model receives a sequence of sentiment data and is trained to reconstruct the same sequence, with the reconstruction loss measured by mean squared error (MSE). Through this process, the Transformer learns to capture the underlying relationships between the sentiment scores of different companies. By capturing these relationships, the Enhancement Transformer can also infer and fill in missing sentiment data that may be absent in the original dataset.

Furthermore, the Enhancement Transformer model was trained using the Adam optimizer with a fixed learning rate of 0.001. Early stopping techniques were not applied because the model demonstrated stable convergence within the predetermined number of training epochs, showing no signs of overfitting.

For the Forecast Transformer model, we trained it to predict stock prices based on both sentiment scores and historical close prices. The model takes sequences of combined sentiment and price information as input and outputs the predicted next-day close prices. Similar to the Enhancement Transformer, MSE is used to measure the prediction loss during training. Through this setup, the model learns to capture how changes in sentiment and past prices jointly influence future stock movements.

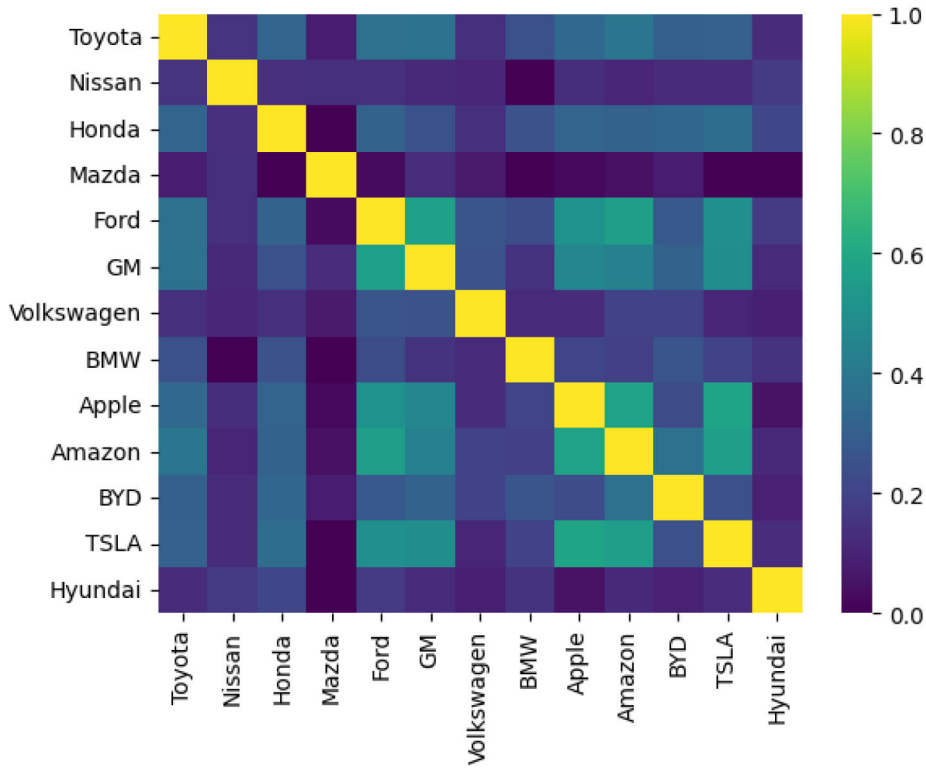
The model was trained using the Adam optimizer with a learning rate of 5×10^{-5}. For each stock dataset, various batch sizes and numbers of training epochs were tested based on validation performance. The batch size typically ranges from 32 to 64, and the number of epochs ranges from 30 to 50.

3.5 Baseline models

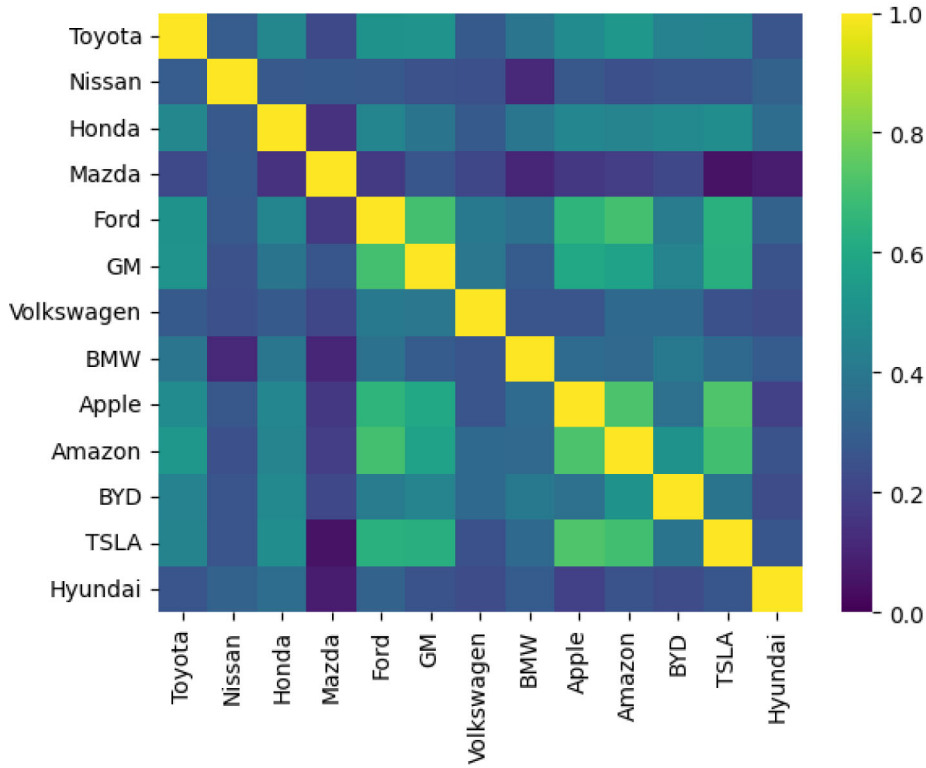
To evaluate the effectiveness of Dual Transformer, Informer, Temporal Fusion Transformer, N-Beats, TCN, LSTM, SVR, and Random Forest are chosen as baseline models for comparison, and the reason to choose these models is because they are used frequently in stock price prediction. For deep learning models such as LSTM and TCN, Optuna Tuner, an open-source hyperparameter optimization framework proposed by [51], can be used to optimize the hyperparameters such as dropout rate, kernel size and number of channels. For machine learning models such as SVR and Random Forest, GridSearchCV can be used to optimize hyperparameters such as kernel function type. We use a sliding window of five days for the historical close prices and corresponding daily sentiment scores as input features. For each time step, the sentiment and close price are concatenated to form a feature vector, resulting in a sequence of five such vectors. This sequence is then fed into models such as LSTM, TCN, SVM, and Random Forest for predicting the stock price of the next day. Informer model shares the same input setup as the Forecast Transformer. The following is a brief introduction to the baseline models.

Informer: Informer, proposed by [32], has mainly three innovative modules compared to the vanilla Transformer model and is designed for prediction of long sequences. The first innovative component is ProbSparse self-attention,

Fig. 7 Comparison of Correlation Differences Before and After Processing



(a) Sentiment Correlation Without Enhancement Transformer Processing



(b) Sentiment Correlation After Enhancement Transformer Processing

which helps reduce time complexity and memory usage. The second innovative component is self-attention distilling, which trims the time dimension of inputs and helps receiving long-sequence input. The third innovative module is called the generative style decoder, which combines historical data of a certain length with a predictive sequence filled with zeros as the input of the decoder. The Informer model has been shown to be able to outperform the Transformer model [28, 46]. In this research, the same Informer structure as in the original paper [32] is applied.

Temporal Fusion Transformer (TFT): TFT is built on a sequence-to-sequence architecture that uses LSTM layers to process time series data. What makes TFT innovative is its use of gated residual networks and variable selection networks, which help the model choose the most relevant inputs at each time step. It also includes a multi-head attention mechanism to focus on important past events, and it separates static, past, and future covariates to better handle complex time series. These design choices allow TFT to provide both strong performance and interpretability. In this study, TFT is implemented using the PyTorch Forecasting library, which supports hyperparameter tuning with Optuna and the integration of additional features such as sentiment scores, making it well suited for our multi-source financial data. Therefore, the TFT hyperparameters are tuned using Optuna.

Neural Basis Expansion Analysis for Time Series Forecasting (N-Beats): N-Beats uses a stack of fully connected blocks, where each block learns to model either trend or seasonality. Each block contains a backcast output, which reconstructs the input, and a forecast output, which predicts future values. The key innovation in N-Beats is this backcast–forecast structure, which allows the model to correct errors over layers and refine its predictions. Unlike models that rely on recurrence or convolutions, N-Beats uses only feed-forward layers, making it fast and easy to train while still capturing complex patterns. We implement N-Beats and hyperparameter tuning using PyTorch Forecasting as well. However, this model only accepts univariate time series (i.e., the target variable's historical values) and does not support additional features such as sentiment or exogenous covariates.

Long Short-Term Memory (LSTM): LSTM is a type of recurrent neural network (RNN) designed to tackle the issue of vanishing or exploding gradients in RNNs. It mainly consists of the Forget Gate, the Input Gate, and the Output Gate, which are used to control the flow of information and retain memory over long periods.

Temporal Convolutional Network (TCN): TCN is a type of convolutional neural network (CNN) designed for sequential modeling. Unlike RNNs, TCN leverages dilated convolutions to capture long-term dependencies in time series data.

Support Vector Regression (SVR): SVR is an extension of the support vector machine (SVM) and can be used to solve the regression problem. SVR works by finding a hyperplane to best fit the data within a defined tolerance margin.

Random Forest: Random Forest is a type of ensemble method. The main logic of Random Forest is to build multiple decision trees and then combine their results based on the majority vote or the average of all trees to improve the accuracy of the prediction.

3.6 Experimental setup

In order to study the effectiveness of using news sentiment of related companies in stock price prediction, experiments are set up in four scenarios, with polarity score of both target and related companies (**with Polarity of Related Firms**), with polarity of target company only (**with Polarity of Target Firm**) and without polarity score (**w/o Polarity Score**). Furthermore, to examine whether changes in the number of related companies impact the model's performance, we also test the models by reducing the number of related companies (**with Fewer Related Firms**). Specifically, we rank the companies based on their Shapley values and select only the top five with the highest scores.

In addition to the polarity score, we also use historical close prices as input to predict the next day's close price. The input features are based on a sliding window of the past 5 days, meaning that the data from the previous 5 days is used to predict the close price for the following day.

4 Experiments

4.1 Related companies selection

Here, we use Toyota as an example to illustrate the analysis and results. First, we used NER to extract the organization names from Toyota news. Table 1 shows the top ten entity names in Toyota news. Then we selected only companies from the entity names, such as Tesla, Ford, and GM. Due to the limitation of the API we used to collect the news data, only publicly listed companies are chosen from the entity names. For each target company, we chose about ten related companies from the extracted entities based on their frequency of occurrence. Table 2 shows the related companies of all target companies. Based on the results, our proposed method can identify not only companies within the same industry or sector as the target company, but also companies from different industries that are connected to the target company. These connections may arise from investor behavior, broader market trends, or other factors that are not apparent to the general public. These connections include relationships such as Apple, Microsoft, Tesla, and IBM for Pepsi, Apple and

Table 1 Top 10 Extracted Entity Names Using NER

Entity Name	Count
NYSE	4721
EV	4220
Tesla	2982
TM	2757
Ford	1766
GM	1591
Reuters	1192
Zacks Investment Research	965
Lexus	955
Free Stock Analysis	880
..	..

Table 2 Related Companies

Target company	Related company
Apple	Google, Amazon, Microsoft, Tesla, Samsung, Nvidia, Meta, Intel, AMD, IBM
Airbus	Boeing, Lockheed Martin, Spirit AeroSystems, Textron, General Electric, Northrop Grumman, Rolls-Royce, Embraer, Raytheon Technologies, Amazon
HSBC	JPMorgan, Standard Chartered, Morgan Stanley, UBS, Citigroup, Goldman Sachs, Credit Suisse, NatWest, Apple, Microsoft
Infusystem	Northwest Pipe, Citizens & Northern, Cabaletta Bio, Luokung Technology, Sanara MedTech, Adicet Bio, Westamerica, CECO Environmental
Tencent	NetEase, Baidu, JD, Alibaba, Microsoft, Apple, Google, Amazon, Meta, Nvidia, Intel
Toyota	Nissan, Honda, Mazda, Ford, GM, Volkswagen, BMW, Tesla, Hyundai, BYD, Apple, Amazon
Pepsi	Coca-Cola, Walmart, Costco, Colgate, Chevron, P&G, Johnson & Johnson, Apple, Microsoft, Amazon, Tesla, IBM
Petrobras	Shell, Braskem, Chevron, ConocoPhillips, ExxonMobil, Cigna Corporation, TotalEnergies, Enbridge, Amazon

Amazon for Toyota, Northwest Pipe, Citizens & Northern, Westamerica, CECO Environmental, and Luokung Technology for InfuSystem.

We also calculated the SHAP values (Sect. 3.2) for the related companies after training the Random Forest model (Figs. 8 and 9). Then we selected the top five companies (Amazon, Apple, BYD, GM, Ford) in this ranking together with close price as input to evaluate the model performances in stock forecasting.

4.2 Stock price prediction

The mean squared error (MSE) is used to evaluate the performance of the models. MSE is the average squared difference between the observed values and the predicted values. A smaller MSE indicates better model performance.

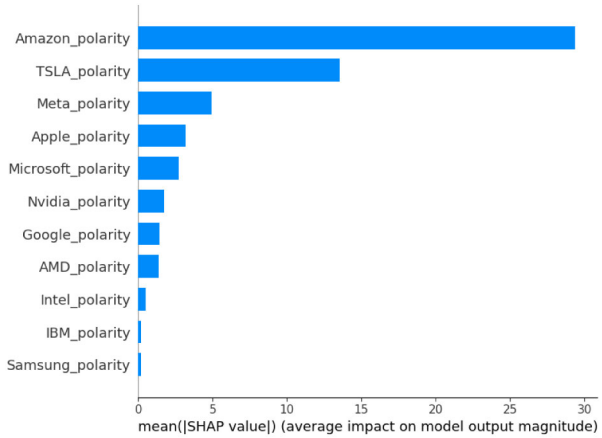
Tables 3, 4, 5, 6, 7, 8, 9 and 10 present the prediction results in terms of MSE for the eight selected companies. All values are expressed as percentages. The proposed Dual Transformer model is applied only in scenarios involving polarity scores of related companies, since the input to the Enhancement Transformer is the polarity score. The MSE values of the Dual Transformer are highlighted in **red** if the Dual Transformer outperforms the other models, which is applied to both scenarios, "with Polarity of Related Firms" and "with Fewer Related Firms".

Based on the results, the main findings are summarized as follows. First, the proposed Dual Transformer consistently outperforms other models across all companies for almost all the time when using features with the polarity of related firms. When using fewer related firms, the Dual Transformer achieves the lowest MSE values in some cases (Tables 6 and 7), indicating that removing companies with minimal contributions to prediction is likely to enhance performance. However, in most cases, the Dual Transformer yields the best result with polarity of more related companies.

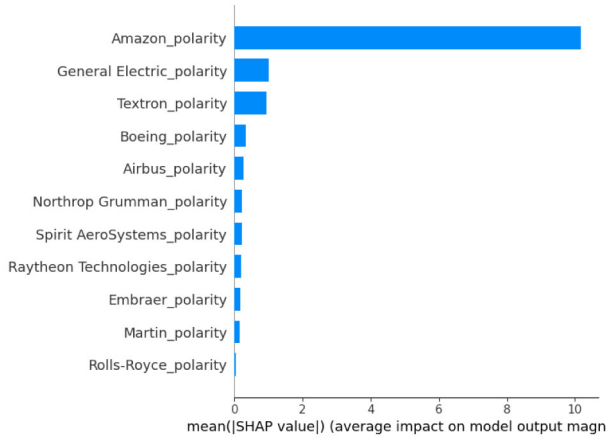
In some instances, the Informer model and TFT with polarity scores as input exhibits comparative results with the Dual Transformer. For example, for HSBC (Table 5), the MSE values of Informer, TFT, and Dual Transformer are closely aligned when using fewer related firms. In addition to the Dual Transformer, both Informer and TFT models show a greater tendency to performance improvement in some cases when polarity scores are added as input (Tables 3, 7, and 10). This indicates that models with a self-attention mechanism, such as Informer and TFT, are better equipped to capture the inter-company relationships that influence stock prices.

N-Beats, while not consistently outperforming attention-based models, shows relatively strong performance in scenarios without polarity scores (e.g., Tables 5 and 10), suggesting its effectiveness in capturing temporal dependencies purely from price history. However, its inability to utilize polarity-based information limits its flexibility in multi-source feature integration compared to Transformer-based approaches.

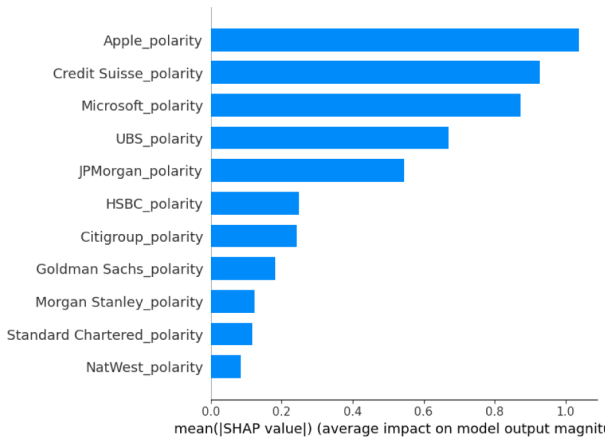
For LSTM, TCN, SVR, and Random Forest, adding polarity scores on top of historical close prices generally does not improve prediction performance. In particular, SVR and Random Forest consistently underperform compared to deep learning models, demonstrating the limitations of traditional



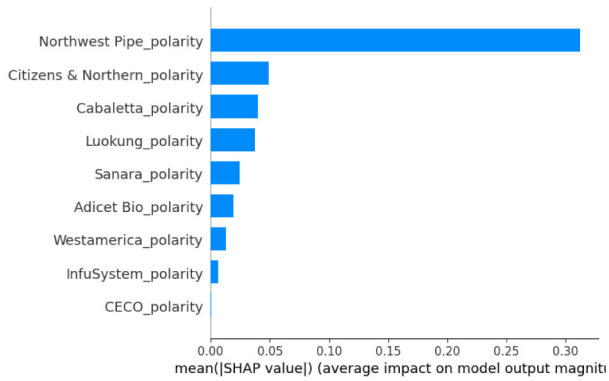
(a) Apple



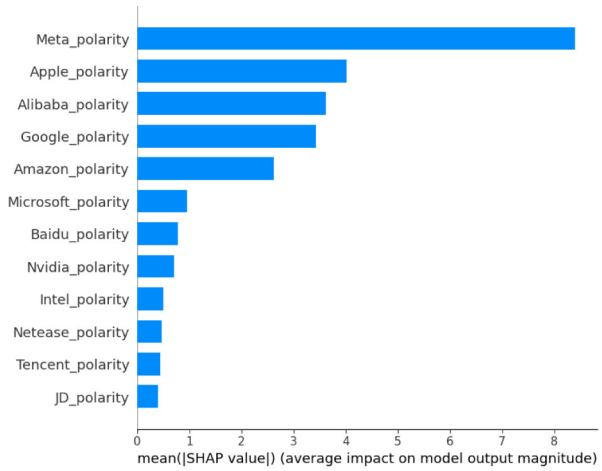
(b) Airbus



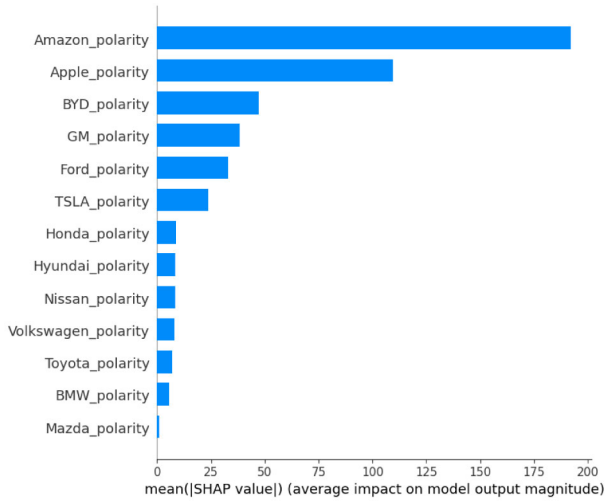
(c) HSBC



(d) Infusystem

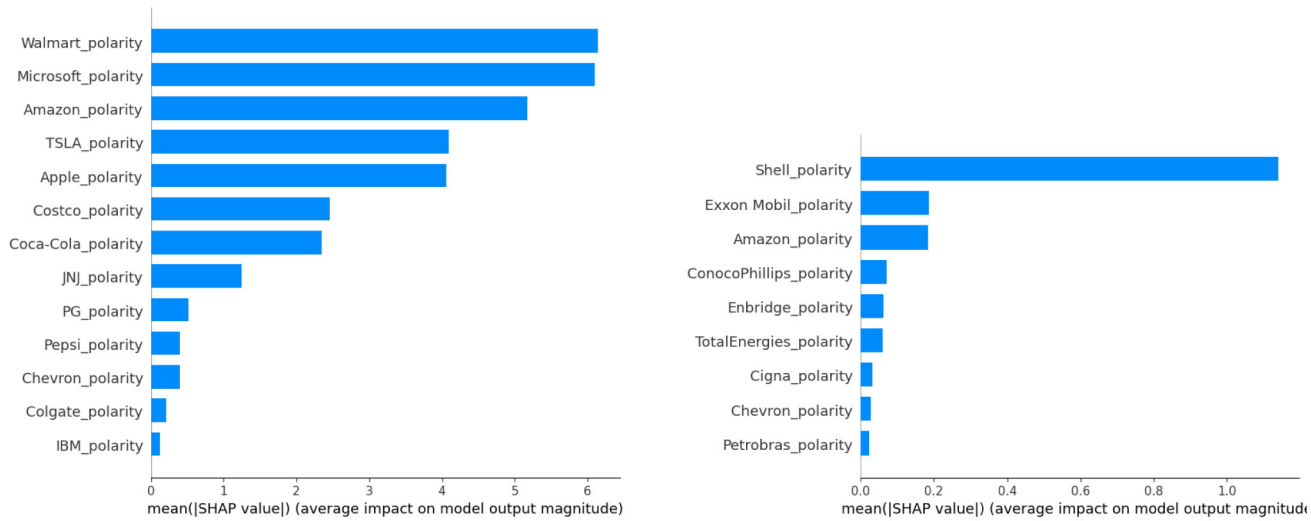


(e) Tencent



(f) Toyota

Fig. 8 Feature Importance Ranking in Terms of SHAP values (1st Part)



(a) Pepsi (b) Petrobras

Fig. 9 Feature Importance Ranking in Terms of SHAP values (2nd Part)

machine learning in capturing the complex and nonlinear relationships in financial time series. Random Forest, in particular, exhibits only trivial performance changes when polarity features are introduced.

Generally, in this experimental setup, the proposed Dual Transformer model exhibits state-of-the-art performance in forecasting stock prices by effectively leveraging the polarity of related firms. Additionally, the results highlight the importance of inter-company relationships and advanced Transformer-based architectures in financial prediction tasks.

4.3 Ablation study

We conducted an ablation study to evaluate the performance of various model configurations on stock price forecasting task for Apple, Pepsi, Toyota, and Tencent. The model configurations include Dual Transformer, Forecast Transformer (FT) alone, Enhancement Transformer (ET) combined with the Informer model, and the Informer model alone. The study is conducted in two feature setups: with polarity of related firms and with fewer related firms. MSE is used for performance evaluation.

Tables 11, 12, 13 and 14 illustrate the results of the ablation study. The MSE values are highlighted in **red** when the dual-structure models outperform the single-structure model. Generally, after adding the ET, the performance of the models tends to improve. The ET combined with Informer achieves competitive results compared to the Dual Transformer. This ablation study also highlights the robustness and effectiveness of the Dual Transformer, particularly in leveraging the polarity of related firms for improved prediction accuracy.

4.4 Further discussion

4.4.1 Related entities filtering using prompt-based LLM

Previously, the selection of related companies from the named entity recognition (NER) output was performed manually. Although this method allows direct human oversight, it is time-consuming, subjective, and may not scale well when applied to larger datasets. To address these limitations, an alternative approach involves using prompt-based large language models (LLMs) to automate the company filtering process. By designing appropriate prompts, LLMs can intelligently identify and extract relevant company entities from unstructured text, reducing the need for manual intervention while improving consistency and scalability. This LLM-based method offers a more systematic and adaptable solution, especially in scenarios where the number of entities is large or continuously evolving.

Table 15 presents a prompt designed for an LLM to extract the top ten related listed companies from a frequency-based entity list. The prompt instructs the model to filter out non-company entities, exclude the target company (Petrobras), consolidate different name or ticker variations referring to the same company, and return the top ten unique firms by frequency. The final output is requested in a structured tabular format for clarity and consistency. Figure 10 shows a sample output generated by the GPT model.

Table 3 Mean Squared Error (%): Apple

Features	Dual transformer	Informer	TFT	N-Beats	TCN	LSTM	SVR	RF
with Polarity of Related Firms	***0.002***	0.032	0.045	–	0.179	0.286	0.568	0.184
with Fewer Related Firms	***0.004***	0.029	0.051	–	0.160	0.233	0.544	0.181
with Polarity of Target Firm	–	0.072	0.076	–	0.152	0.158	0.404	0.180
w/o Polarity Score	–	0.071	0.088	0.070	0.142	0.150	0.372	0.178

Table 4 Mean Squared Error (%): Airbus

Features	Dual transformer	Informer	TFT	N-Beats	TCN	LSTM	SVR	RF
with Polarity of Related Firms	***0.018***	0.035	0.062	–	0.247	0.295	0.143	0.082
with Fewer Related Firms	***0.032***	0.034	0.066	–	0.268	0.281	0.135	0.082
with Polarity of Target Firm	–	0.051	0.058	–	0.105	0.079	0.153	0.082
w/o Polarity Score	–	0.059	0.061	0.039	0.087	0.084	0.077	0.082

Table 5 Mean Squared Error (%): HSBC

Features	Dual transformer	Informer	TFT	N-Beats	TCN	LSTM	SVR	RF
with Polarity of Related Firms	***0.022***	0.041	0.037	–	0.091	0.078	0.246	0.042
with Fewer Related Firms	0.041	0.049	0.043	–	0.060	0.059	0.246	0.043
with Polarity of Target Firm	–	0.035	0.041	–	0.082	0.043	0.114	0.043
w/o Polarity Score	–	0.030	0.038	0.027	0.047	0.049	0.114	0.043

Table 6 Mean Squared Error (%): Infusystem

Features	Dual transformer	Informer	TFT	N-Beats	TCN	LSTM	SVR	RF
with Polarity of Related Firms	***0.032***	0.083	0.064	–	0.156	0.115	0.342	0.093
with Fewer Related Firms	***0.029***	0.038	0.055	–	0.333	0.147	0.329	0.092
with Polarity of Target Firm	–	0.040	0.058	–	0.081	0.089	0.113	0.092
w/o Polarity Score	–	0.067	0.061	0.041	0.083	0.081	0.113	0.092

Table 7 Mean Squared Error (%): Pepsi

Features	Dual Transformer	Informer	TFT	N-Beats	TCN	LSTM	SVR	RF
with Polarity of Related Firms	***0.026***	0.065	0.044	–	0.298	0.953	0.205	0.136
with Fewer Related Firms	***0.022***	0.028	0.039	–	0.150	0.143	0.252	0.136
with Polarity of Target Firm	–	0.077	0.069	–	0.376	0.153	0.210	0.138
w/o Polarity Score	–	0.088	0.062	0.056	0.126	0.129	0.217	0.137

Table 8 Mean Squared Error (%): Petrobras

Features	Dual Transformer	Informer	TFT	N-Beats	TCN	LSTM	SVR	RF
with Polarity of Related Firms	***0.023***	0.056	0.041	–	0.304	0.207	0.083	0.048
with Fewer Related Firms	***0.031***	0.059	0.048	–	0.180	0.233	0.083	0.048
with Polarity of Target Firm	–	0.350	0.067	–	0.043	0.045	0.082	0.048
w/o Polarity Score	–	0.041	0.052	0.208	0.046	0.049	0.082	0.048

Table 9 Mean Squared Error (%): Toyota

Features	Dual Transformer	Informer	TFT	N-Beats	TCN	LSTM	SVR	RF
with Polarity of Related Firms	***0.009***	0.027	0.031	–	0.137	0.171	0.181	0.050
with Fewer Related Firms	***0.010***	0.026	0.038	–	0.106	0.134	0.187	0.049
with Polarity of Target Firm	–	0.016	0.024	–	0.047	0.121	0.187	0.049
w/o Polarity Score	–	0.020	0.029	0.031	0.038	0.044	0.083	0.047

Table 10 Mean Squared Error (%): Tencent

Features	Dual Transformer	Informer	TFT	N-Beats	TCN	LSTM	SVR	RF
with Polarity of Related Firms	***0.028***	0.047	0.043	–	0.315	0.413	0.203	0.100
with Fewer Related Firms	0.035	0.106	0.063	–	0.245	0.321	0.201	0.099
with Polarity of Target Firm	–	0.041	0.059	–	0.103	0.107	0.201	0.099
w/o Polarity Score	–	0.054	0.056	0.033	0.095	0.100	0.201	0.099

Table 11 Ablation Study, MSE (%): Apple

Features	Dual transformer	FT	ET + Informer	Informer
with Polarity of Related Firms	***0.002***	0.016	***0.003***	0.032
with Fewer Related Firms	***0.004***	0.017	***0.025***	0.029

Table 12 Ablation Study, MSE (%): Pepsi

Features	Dual transformer	FT	ET + Informer	Informer
with Polarity of Related Firms	***0.026***	0.035	***0.037***	0.061
with Fewer Related Firms	***0.022***	0.028	***0.038***	0.044

Table 13 Ablation Study, MSE (%): Toyota

Features	Dual transformer	FT	ET + Informer	Informer
with Polarity of Related Firms	***0.009***	0.021	***0.026***	0.027
with Fewer Related Firms	***0.010***	0.022	***0.023***	0.026

Table 14 Ablation Study, MSE (%): Tencent

Features	Dual transformer	FT	ET + Informer	Informer
with Polarity of Related Firms	***0.028***	0.044	***0.039***	0.047
with Fewer Related Firms	***0.030***	0.034	***0.022***	0.106

Table 15 Sample Prompt for Extracting Related Companies

Given a list of entities with their corresponding frequency counts {**Entity List Input**}, perform the following steps to extract the top 10 listed companies

1. Filter the data to include only actual listed companies. Exclude entities such as financial research firms (e.g., Zacks), stock exchanges (e.g., NYSE, NASDAQ), media sources (e.g., Reuters, Bloomberg), and general terms (e.g., EPS, Offer Documents)
2. Exclude Petrobras (ticker: PBR) (Target Company) from the output, even if it appears among the top companies
3. Deduplicate company entries by consolidating different representations of the same company. For instance, merge: *Chevron*, *CVX*, *Chevron Corporation*; *Exxon*, *XOM*, *Exxon Mobil Corporation*, *ExxonMobil*; *ConocoPhillips*, *COP*; *Moody's*, *MOODY*, *Moody's Corporation*
4. After cleaning and deduplication, rank the companies by total count and output the top 10 unique companies along with their final frequencies
5. Present the final result in a `tabular` environment showing: Rank, Company Name (consolidated), and Total Count

Fig. 10 Output Sample of GPT Model

✅ **Output Format**

Rank	Company Name	Consolidated Variants	Total Count
1	ExxonMobil	Exxon, XOM, ExxonMobil, Exxon Mobil Corporation, Exxon Mobil	**477**
2	Chevron	Chevron, CVX, Chevron Corporation	**343**
3	ConocoPhillips	ConocoPhillips, COP	**294**
4	Shell	Shell	**214**
5	Braskem	Braskem	**183**
6	Cigna Corporation	Cigna Corporation, CI	**202**
7	TotalEnergies	TotalEnergies	**112**
8	Enbridge	Enbridge	**64**
9	Amazon	Amazon	**64**

4.4.2 Event study

In this section, we use a method called event study to further illustrate how the relationships between companies captured by the Enhancement Transformer match the real-world situations.

In finance, the event study can help identify companies whose cumulative abnormal returns follow a pattern similar to that of the target company. This similarity suggests that the events may have comparable impacts on these companies, indicating a potential connection between them.

The company relationships identified through event studies can be used for comparison with those captured by the Enhancement Transformer. The details of the event study will be explained below, followed by a comparison of the relationships identified by the event study and the Enhancement Transformer.

In finance, event study is often used to estimate how asset prices react to announcements of events that contain new information relevant to the value of the underlying assets. To conduct an event study, the main idea is to compare the asset price that occurred as a consequence of the announcement of the event with a hypothetical asset price that would have occurred if no event had happened. The former is called the realized return (return calculated using the stock prices in the event window), and the latter is called the expected return (return calculated using the stock prices in the estimation period). That is, we want to calculate the difference between the realized and expected return, called the abnormal return (Equation 7). The expected return is calculated using the market model (Equation 8). The final step is to calculate the cumulative abnormal return (CAR) (Equation 9). Companies that show similar patterns in CAR during the event study are likely connected. Additionally, if the CARs of some companies frequently move in opposite directions, it may suggest a relationship where favorable news for one translates to unfavorable news for another. For example, negative news for one company could translate into positive news for its competitors.

The event study must specify the event date (business day right after the event announcement), the estimation period (period to calculate the expected return), and the event window (period to calculate the realized return). Here, we used before and after 10 days surrounding the event date as the event window and the last 130 days to last 11 days before the event as the estimation period (Fig. 11). The length of the estimation period is arbitrary, but in practice, the length of the estimation period typically ranges from 100 to 250 trading days.

$$AR_{i,t} = R_{i,t} - ER_{i,t}, \quad t \in [-10, 10] \tag{7}$$

where $AR_{i,t}$ is the abnormal return for the stock i on day t, $R_{i,t}$ is the realized return for the stock i on day t, calculated using the stock prices in the event window, and $ER_{i,t}$ is the expected return for the stock i on day t, which is calculated using Equation 8.

$$ER_{i,t} = \alpha_i + \beta_i R_{m,t} + \epsilon_{i,t} \tag{8}$$

where parameters such as α_i and β_i can be obtained using the stock data in the estimation period $t \in [-130, -11]$. α_i is the intercept term, β_i is the slope coefficient, representing the sensitivity of the return of the stock i to the market

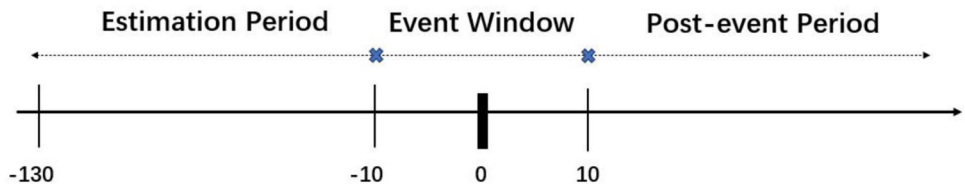


Fig. 11 Event Study

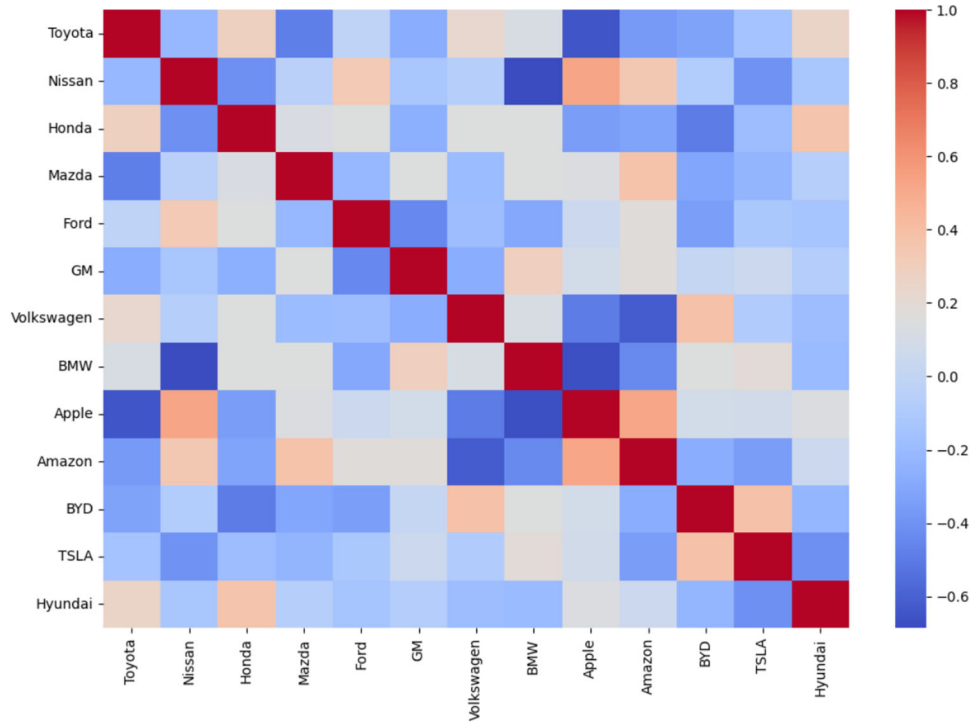


Fig. 12 Attention Scores between Companies with ET Processing (Different From Fig. 7b)

return, $R_{m,t}$ is the return of the market index on day t, $\epsilon_{i,t}$ is the error term. Examples of market index include S&P 500 (US), Hang Seng Index (Hong Kong), Taiwan Capitalization Weighted Stock Index (Taiwan), Swiss Market Index (Switzerland), and Korea Composite Stock Price Index (South Korea). Stocks from different countries are associated with the market index of their respective exchanges. For example, Tencent, listed on the Hong Kong Stock Exchange, uses the Hang Seng Index for calculations.

$$CAR_{i,t} = \sum_{t'=-10}^{t} AR_{i,t'}, \tag{9}$$

where $CAR_{i,t}$ is the cumulative abnormal return for the stock i during the event window, $AR_{i,t'}$ is the abnormal return for the stock i on day t'.

4.4.3 Analysis of enhancement transformer with event study results

Here, we use related companies of Toyota as an example. Figure 13 shows the results of the event study using Toyota's scandal on 3 June 2024. On this date, Toyota acknowledged irregularities in safety tests conducted by its subsidiaries, Daihatsu and Hino Motors. Figure 14 shows the event study using the Russia-Ukraine War, which started on 24 February, 2022. These two figures reveal changes in cumulative abnormal returns for companies such as BYD, Nissan, Volkswagen, and Honda after these events, which indicate relationships between Toyota and other companies.

Figure 12 uses attention scores from the transformer model (second layer) to measure the interaction between companies based on their polarity scores. The attention scores here are computed using data from 2022 to 2024, which covers the event dates mentioned above. Attention scores are used to represent the importance assigned to each input feature when making predictions, providing information on the relationships between companies. By aggregating and analyzing these scores, we can quantify the degree to which companies are related within the learned representation of the transformer. In this research, we used two layers and four attention heads in the calculation of self-attention. The attention scores generated by each layer represent the combined contributions from all heads (Equation 2), with the attention scores averaged across all heads.

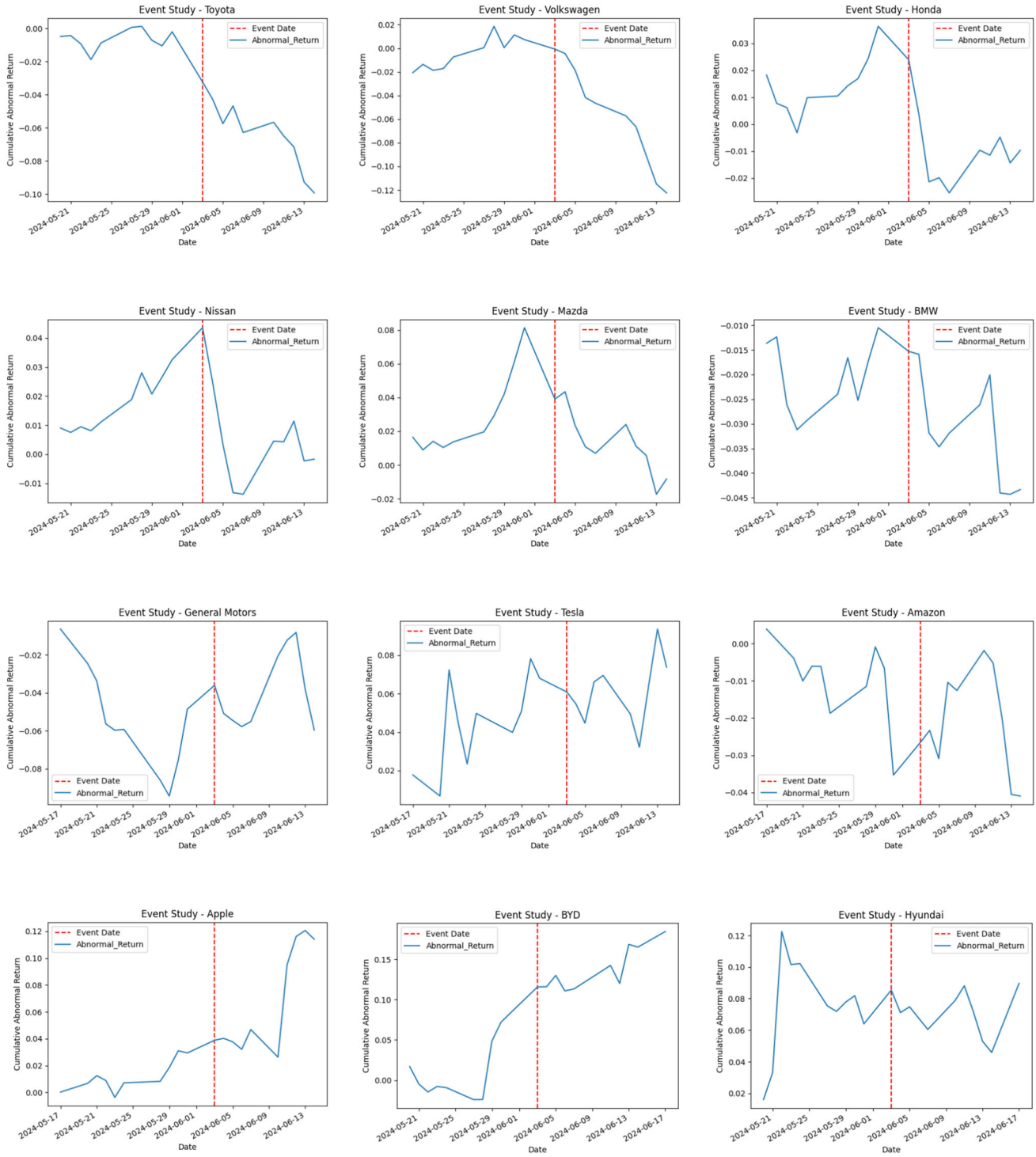


Fig. 13 Event Study - Toyota Scandal: Related Companies of Toyota

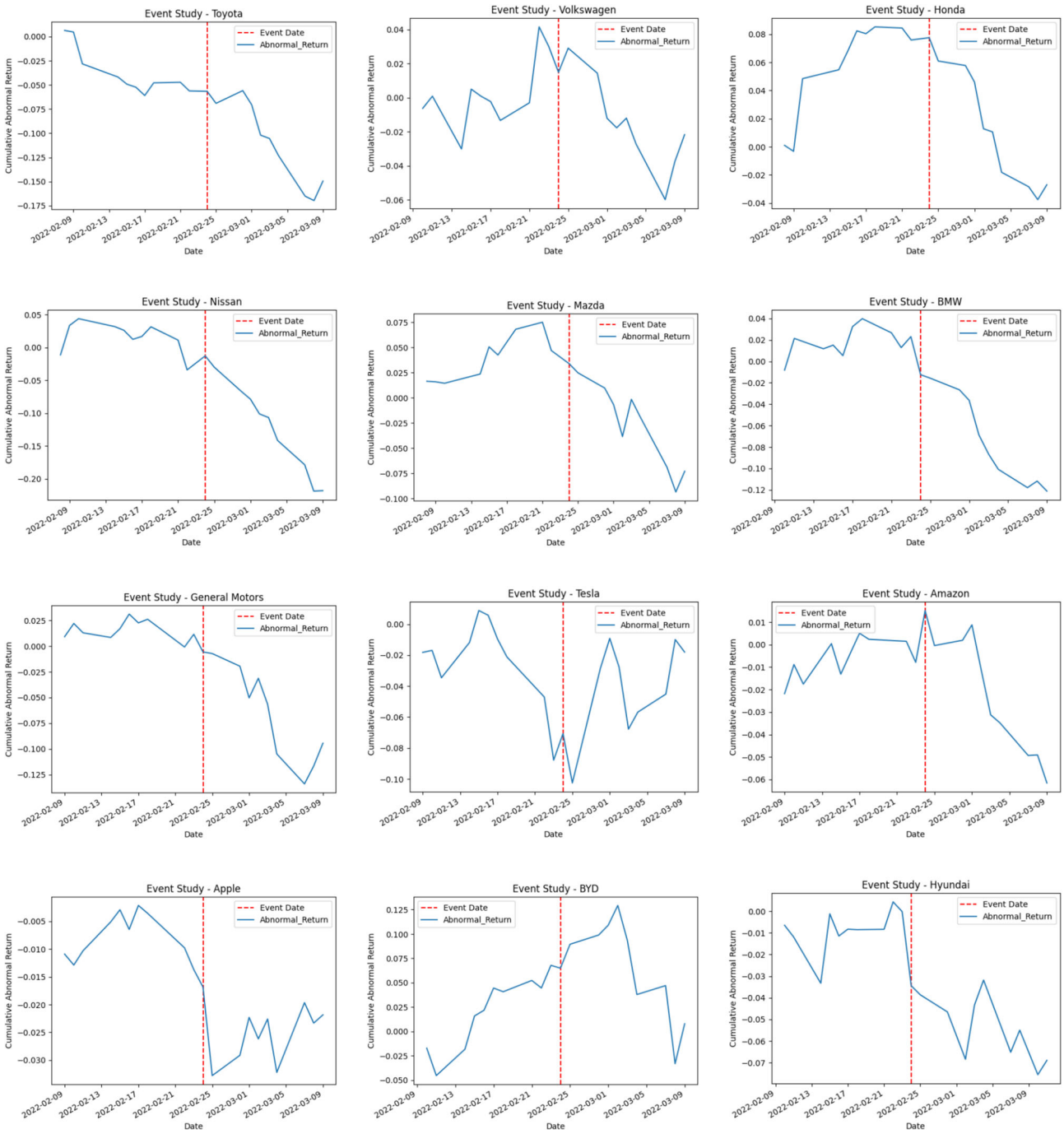


Fig. 14 Event Study - Russia–Ukraine War: Related Companies of Toyota

According to Fig. 13, after Toyota's scandal, the abnormal returns of most of auto companies dropped. Although the magnitude of the abnormal returns varied, their overall direction or trend was largely consistent. In particular, Honda, Nissan, and Mazda exhibited remarkably similar movements in their abnormal returns. Similarly, Ford, General Motors, and BMW showed similar patterns, as did Toyota and Volkswagen. These relationships are also captured in the transformer results (Fig. 12). For example, Toyota shows a relatively strong positive interaction with Honda and Volkswagen, suggesting that their sentiment trends are closely related. Companies such as Volkswagen and BMW exhibit a delayed or less pronounced reaction compared to Japanese automakers such as Nissan and Honda, potentially reflecting regional market differences or varying levels of exposure to the scandal.

Furthermore, based on Fig. 12, Tesla stands out with predominantly negative or weak interaction with other companies, such as Toyota, Mazda, Honda, and Ford, which is consistent with its unique pattern in the event study (Figs. 13 and 14). This reflects the reality that Tesla focuses primarily on the electric vehicle market, while others remain more diversified in traditional and hybrid vehicles.

BYD and Tesla exhibit positive interactions and relatively similar patterns in the event study, highlighting their shared focus on the electric vehicle market.

Apple displays weak but noticeable interactions with Tesla in some aspects, reflecting their alignment with innovation and technology-driven sentiment, even though they operate in different industries.

In general, based on the results and analysis, the proposed transformer model can effectively capture and enhance the relationships and interactions between companies. In addition, the consistency between some results of the event study and the transformer demonstrates the transformer's ability to uncover meaningful relationships between entities, complementing traditional event study analyses.

5 Conclusion and limitations

This paper proposed a novel sentiment correlation-based method and a Dual Transformer model to forecast the stock prices of eight companies. Some key findings of this paper are as follows. (1) The Enhancement Transformer (encoder transformer) of our proposed Dual Transformer model can help increase the correlation strength of the companies' polarity scores. (2) The proposed Dual Transformer model with sentiments of related companies as input can outperform models like Informer and LSTM. (3) Models with attention mechanisms, such as Informer, are more likely to improve prediction performance when sentiment data is added as input, which demonstrates the strength of self-attention mechanisms in capturing inter-company relationships, which are crucial for understanding the influence of related firms on stock prices.

This research has certain limitations. The API used to collect news data is limited to publicly listed companies, with a bias toward American companies, resulting in less comprehensive coverage for other regions. To address this, alternative data sources will be explored in future work.

Another limitation lies in the NER process used to extract companies mentioned in financial news. Although NER effectively identifies potential companies, the final selection of relevant companies is performed manually on the basis of frequency, which introduces subjectivity and may limit scalability, especially for large datasets. This manual, frequency-based selection may also capture media salience rather than genuine economic or financial linkages, potentially impacting the accuracy of modeling inter-company sentiment spillovers. More robust, data-driven approaches such as graph community detection, price correlation networks, or industry classification proximity offer promising alternatives and will be investigated in future studies to improve model robustness and generalizability.

Author Contributions Qizhao Chen conceptualized the idea, conducted the experiments, and drafted the manuscript. Hiroaki Kawashima contributed to the conceptualization of the idea and provided critical revisions to the manuscript.

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Data Availability The news datasets used in this study were collected using the News Feed and News Sentiment data API. The stock data were collected using the Python library yfinance.

Declarations

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Materials availability Not applicable.

Code availability GitHub Repository

Conflict of interest The authors declare no Conflict of interest.

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